

7. $\begin{bmatrix} 1 & 4 & -3 & 0 \\ -2 & -1 & 5 & 1 \\ -4 & -5 & 7 & 5 \end{bmatrix}$ No. of columns = 4 > No. of entries in each column = 3

By Theorem 8, the columns are linearly dependent.

8. In Exercises 9 and 10, (a) for what values of h is v_3 in $\text{span}\{v_1, v_2\}$, and (b) for what values of h is $\{v_1, v_2, v_3\}$ linearly dependent? Justify each answer.

9. $v_1 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, v_2 = \begin{bmatrix} -3 \\ 9 \\ -6 \end{bmatrix}, v_3 = \begin{bmatrix} 5 \\ -7 \\ h \end{bmatrix}$

3. Solution: Suppose v_3 is in $\text{Span}\{v_1, v_2\}$. By defn, v_3 is a linear combination of v_1 and v_2 . $\therefore \exists c_1, c_2 \in \mathbb{R}$ s.t. $c_1 v_1 + c_2 v_2 = v_3$. By Theorem 1,

$\{v_1, v_2, v_3\}$ is linearly dependent.

$c_1 \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 9 \\ -6 \end{bmatrix} = \begin{bmatrix} 5 \\ -7 \\ h \end{bmatrix}$ By defn. of matrix equation $Ax=b$, we get: $\begin{bmatrix} 1 & -3 \\ -3 & 9 \\ 2 & -6 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 5 \\ -7 \\ h \end{bmatrix}$

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5. From Theorem 3, the given matrix equation has the same solution set as the system of equations corresponding to the augmented matrix $\begin{bmatrix} 1 & -3 & 5 \\ -3 & 9 & -7 \\ 2 & -6 & h \end{bmatrix}$

$\begin{bmatrix} 1 & -3 & 5 \\ -3 & 9 & -7 \\ 2 & -6 & h \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 3 \times \text{Row 1}}]{\text{Row 3} = \text{Row 3} \\ - 2 \times \text{Row 1}} \begin{bmatrix} 1 & -3 & 5 \\ 0 & 0 & 8 \\ 2 & -6 & h \end{bmatrix} \xrightarrow[\substack{\text{Row 3} = \text{Row 3} \\ - 2 \times \text{Row 1}}]{\text{Row 3} = \text{Row 3} \\ - 2 \times \text{Row 1}} \begin{bmatrix} 1 & -3 & 5 \\ 0 & 0 & 8 \\ 0 & 0 & h-10 \end{bmatrix}$

Verification of whether the final matrix is in echelon form.

1. Row 1 and Row 2 are non-zero rows. Row 3 is either a non-zero row ($h \neq 10$) or a zero row ($h = 10$). $\therefore$ The condition that all non-zero rows must be above the zero rows is True.

2. Row 1 leading element = 1 in Column 1
Row 2 leading element = 8 in Column 3

$\therefore$ The condition that for each row, the leading element must be in a column to the right of the columns of the leading elements of all rows above it is True. $\therefore h-10=0 \Rightarrow h=10$. (For Cond. 2 to hold)